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Project 01 : Advanced User Registration Form using React

Aim

To design and develop an Advanced User Registration Form using React that validates username, Aadhaar name, email, phone number, password, confirm password, and photo upload.

Procedure

- Create a React application using Vite.
- Design a form component with multiple input fields.
- Use useState hook to store form data.
- Implement input fields like username, Aadhaar name, email, phone number, password, confirm password, and photo.
- Add validation rules:
 - Username and Aadhaar name must be the same.
 - Phone number must be exactly 10 digits.
 - Email must contain '@'.
 - Password must be at least 6 characters.
 - Confirm password must match the password.
 - Handle photo file upload using the file input.
- Implement handleChange function to update form data dynamically.
- Implement validate() function to check all conditions.
- Use handleSubmit() to prevent default submission and validate the form.
- Apply CSS styling for better UI.
- Run the application and test different inputs.

React Concepts Used

- Functional Component is used to create the form UI.
- useState Hook is used to store and manage form data.
- Controlled Components are used to handle input fields with state.
- Event Handling (onChange, onSubmit) is used for user interactions.
- Conditional Checking is used for validation of inputs.
- Form Handling is implemented using preventDefault() to avoid page reload.
- File Input Component is used for photo upload.
- Dynamic State Update is used to update form values in real-time.

Code

App.jsx

```
import React from "react";
import Form from "../Form";
import "../App.css";
```

```
function App() {
  return (
    <div>
      <Form />
    </div>
  );
}
```

export default App;

Form.jsx

```
import React, { useState } from "react";
import "../Form.css";
```

```
function Form() {
  const [form, setForm] = useState({
    username: "",
    aadhaar: "",
    email: "",
    phone: "",
    password: "",
    confirmPassword: "",
    photo: null
  });
```

```
  const handleChange = (e) => {
    const { name, value, files } = e.target;
```

```
    if (name === "photo") {
      setForm({ ...form, photo: files[0] });
    } else {
      setForm({ ...form, [name]: value });
    }
  };
};
```

```
  const validate = () => {
    if (form.username !== form.aadhaar) {
      alert("Username and Aadhaar Name must be same");
      return false;
    }
  };
}
```

```

    }

    if (form.phone.length !== 10 || isNaN(form.phone)) {
        alert("Phone must be 10 digits");
        return false;
    }

    if (!form.email.includes("@")) {
        alert("Invalid email");
        return false;
    }

    if (form.password.length < 6) {
        alert("Password must be at least 6 characters");
        return false;
    }

    if (form.password !== form.confirmPassword) {
        alert("Passwords do not match");
        return false;
    }

    return true;
};

const handleSubmit = (e) => {
    e.preventDefault();

    if (validate()) {
        alert("Form Submitted Successfully");
    }
};

return (
    <form onSubmit={handleSubmit}>
        <input name="username" placeholder="Username" onChange={handleChange} required />
        <input name="aadhaar" placeholder="Aadhaar Name" onChange={handleChange} required />
        <input name="email" placeholder="Email" onChange={handleChange} required />
        <input name="phone" placeholder="Phone" onChange={handleChange} required />
        <input type="password" name="password" placeholder="Password"
    onChange={handleChange} required />
        <input type="password" name="confirmPassword" placeholder="Confirm Password"
    onChange={handleChange} required />
        <input type="file" name="photo" onChange={handleChange} />
    </form>
);

```

```

        <button type="submit">Submit</button>
    </form>
);
}

export default Form;
Form.css
body {
    margin: 0;
    font-family: Arial, sans-serif;
    background: linear-gradient(120deg, #667eea, #764ba2);
    min-height: 100vh;
}

.container {
    display: flex;
    justify-content: center;
    align-items: center;
    padding: 50px 0;
}

form {
    background: rgba(255, 255, 255, 0.95);
    padding: 30px;
    border-radius: 20px;
    width: 380px;
    box-shadow: 0 10px 30px rgba(0, 0, 0, 0.2);
}

h2 {
    text-align: center;
    color: #4b3f72;
    margin-bottom: 25px;
}

input,
textarea,
select {
    width: 100%;
    padding: 11px;
    margin: 8px 0 15px;
    border: 1px solid #ccc;
    border-radius: 8px;
    box-sizing: border-box;
    font-size: 14px;
    outline: none;

```

```
}

input:focus,
textarea:focus,
select:focus {
  border-color: #764ba2;
  box-shadow: 0 0 5px rgba(118, 75, 162, 0.3);
}

textarea {
  height: 90px;
  resize: none;
}

.phone {
  display: flex;
  gap: 10px;
}

.phone input {
  flex: 1;
}

.buttons {
  display: flex;
  justify-content: center;
  gap: 15px;
  margin-top: 10px;
}

button {
  padding: 11px 25px;
  border: none;
  border-radius: 8px;
  background: #764ba2;
  color: white;
  font-size: 14px;
  cursor: pointer;
  transition: 0.3s;
}

button:hover {
  background: #5a367f;
  transform: translateY(-2px);
}
```

Output

Username

Aadhaar Name

Email

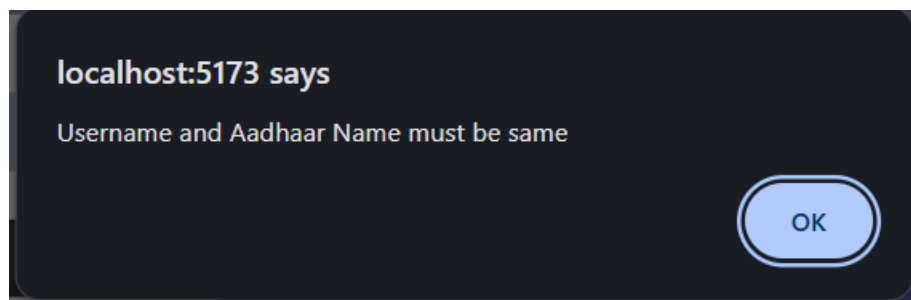
Phone

Password

Confirm Password

Choose file No file chosen

Submit



Result

The Advanced User Registration Form was successfully developed using React. The project demonstrates useState, controlled form inputs, event handling, input validation, file input handling, and CSS styling.